

Data Analyst – Excel & VBA Assignment

(Revenue, Commission & Settlement Analytics)

Business Context (Must Read Carefully)

You are given a transactional earnings file representing payments processed through Edviron's platform.

There are **three pricing layers** involved in every transaction:

1. **Edviron Buying Price**
→ Rate at which Edviron buys payment processing (cost to Edviron)
2. **Partner Pricing (ERP Pricing)**
→ Rate at which Edviron sells payment processing to ERP partners
3. **Merchant Pricing (School Pricing)**
→ Rate at which ERP sells payment processing to schools

Revenue Definitions

- **ERP Revenue**
= Merchant Pricing – Partner Pricing
- **Edviron Net Revenue**
= Partner Pricing – Edviron Buying
- **Edviron Gross Revenue**
= ERP Revenue + Edviron Net Revenue
(Note: Payment Gateway settles the **entire gross amount to Edviron**, and Edviron later settles ERP's share)

Your task is to **analyze, compute, report, and visualize** these revenues using **Excel and VBA**.

Dataset Provided

You will receive:

- A raw Excel file containing transactional data with fields such as:
 - Order ID
 - Transaction Amount
 - Payment Method
 - Bank
 - Gateway
 - Student Name
 - Vendor Amount
 - Partner Pricing (Flat / %)
 - Merchant Pricing (Flat / %)
 - Edviron Buying (Flat / %)
 - Status (SUCCESS / FAILED / PENDING)
 - Capture Status
 - ERP Commission

Assume this is raw, unclean data.

Assignment Objectives

Part 1: Data Cleaning & Normalization (Excel)

1. Clean the dataset:
 - Handle missing values

- Standardize pricing formats (Flat vs Percentage)
 - Ensure numeric consistency
 - 2. Convert **all pricing fields into absolute INR values per transaction**, regardless of whether the input is Flat or Percentage.
 - 3. Create helper columns where required, but maintain clarity and auditability.
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Part 2: Revenue & Commission Computation

For each transaction, compute:

1. **ERP Revenue**
2. **Edviron Net Revenue**
3. **Edviron Gross Revenue**
4. **Gateway Fees (if inferable)**
5. **Net Settlement Amounts**
 - Amount payable to ERP
 - Amount retained by Edviron

Clearly document assumptions wherever data is missing or implicit.

Part 3: Excel Reports (No VBA Yet)

Create **separate structured reports** (preferably on different sheets):

1. **Daily / Weekly / Monthly Revenue Summary**
 - Total Transaction Volume
 - ERP Revenue

- Edviron Net Revenue
- Edviron Gross Revenue
- Unique Users

2. Partner (ERP-wise) Performance

- Transactions
- Revenue split (ERP vs Edviron)
- Average take rate

3. Gateway-wise & Payment Method Analysis

- Volume
- Revenue
- Success vs Failure rates

4. Pending vs Settled Exposure Report

- Capture Pending
- ERP payable outstanding
- Edviron receivable risk

Use Pivot Tables wherever appropriate.

Part 4: Excel VBA Dashboard (Core Evaluation)

Build an **interactive Excel dashboard using VBA**, not Power BI.

Dashboard must include:

Key Metrics (Top Tiles)

- Total Transactions

- Total GMV
- ERP Revenue
- Edviron Net Revenue
- Edviron Gross Revenue
- Pending Exposure
- Unique Users
- Frequency of payment of same user

Visuals (Charts)

- Revenue split (Edviron vs ERP)
- Time-based trend (Daily / Monthly)
- Gateway-wise contribution
- Payment method mix

Controls (VBA Mandatory)

- Date Range Selector
- Partner (ERP) Filter
- Gateway Filter
- Payment Method Filter

All visuals and metrics should dynamically update using **VBA-driven logic**, not manual filtering.

Part 5: VBA Expectations

- Use VBA for:

- Dynamic filtering
 - Metric recalculation
 - Dashboard refresh button
 - Code should be:
 - Modular
 - Commented
 - Readable
 - No hard-coded cell references for business logic.
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Deliverables

1. **Excel File (.xlsx / .xlsm)**
 - Clean data
 - Computation sheets
 - Reports
 - VBA Dashboard
 2. **1–2 page Explanation Note (PDF or Sheet)**
 - Revenue logic explanation
 - Assumptions made
 - Dashboard design rationale
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Evaluation Criteria

Area	Weight
Financial logic accuracy	High
Excel modeling & pivots	High
VBA quality & usability	Very High
Dashboard clarity	High
Business thinking	High
Documentation	Medium

Bonus (Not Mandatory, But Strong Signal)

- Automated reconciliation logic
- Exception flagging (negative margins, pricing errors)
- Scenario toggles (Flat vs % stress test)
- Clean UX/UI polish